

Today we will cover

#Layers of BV

#Arteriosclerosis

#Vasculitis

🩸 Blood vessel walls are organized into three concentric layers: the tunica intima, the tunica media, and the tunica adventitia. These layers are present in all vessels but are most distinct in larger vessels, particularly arteries. ❤️

1. Tunica Intima (Inner Layer) 🔬

The intima is the innermost layer in direct contact with the blood. 🩸 It consists of:

🌻 Endothelial Cells 🚚: A continuous monolayer of cells that provides a non-thrombogenic lining.

👤 Basement Membrane 🧱: A thin layer of connective tissue supporting the endothelial cells.

★ Internal Elastic Lamina 🟡: A dense elastic membrane that separates the intima from the media.

2. Tunica Media (Middle Layer) 💪

The media is primarily responsible for the structural integrity of the vessel and the regulation of blood flow. 🔄 Its composition includes:

🌍 Smooth Muscle Cells (SMCs) 🌀: These cells are circularly or spirally arranged and mediate vasoconstriction and vasodilation.

Extracellular Matrix (ECM) 📦: Includes collagen, elastin, and proteoglycans.

💥 Variations by Vessel Type 🩸:

In large elastic arteries (like the aorta) ❤️ elastic fibers alternate with SMCs to allow for expansion and recoil.

In medium-sized muscular arteries 🟡 the media is composed almost entirely of SMCs.

❄️ External Elastic Lamina 🟡: In some arteries, this membrane defines the transition between the media and the adventitia.

3. Tunica Adventitia (Outer Layer) 🟣

The adventitia, also known as the tunica externa, is the outermost layer of the vessel wall.

🌐 It contains:

👉 Connective Tissue 📦: Loose tissue that anchors the vessel to surrounding structures.

👉 Vasa Vasorum 🚚: Literally “vessels of the vessels,” these are small arterioles that perfuse the outer two-thirds of the media in large vessels, as diffusion from the lumen is insufficient.

👉 Nerves and Lymphatics ⚡🟢: Nerve fibers (nervi vasorum) and lymphatic channels are also present in this layer.

Structural Differences Across the Vasculature 🌐

The thickness and composition of these layers vary depending on the hemodynamic demands of the specific vessel type. 📊

👉 Arteries 📌: Have thicker walls than veins, invested with several reinforcing layers of SMCs to accommodate high blood pressure and pulsatile flow. 💥

👉 Veins 📌: Are distensible and thin-walled with less distinct layers, adapted to the lower pressures on the venous side of the circulation. 🌊

👉 Capillaries 📌: Lacking a media and adventitia, they consist essentially of a single-cell lining of endothelial cells on a basement membrane to facilitate the rapid exchange of nutrients and gases. 🔄

💪 VASCULAR RESPONSE TO INJURY

💧 The vascular response to injury is the fundamental basis for most vascular disorders, ranging from atherosclerosis to hypertension-related changes. This response follows a hierarchical and conceptual framework where the endothelial cell acts as the primary sensor and responder, and the smooth muscle cell executes the structural repair, which can eventually lead to luminal occlusion if the stimulus is persistent. ⚠️

I. Fundamental Concept: Intimal Thickening 🧱

Intimal thickening is the stereotypical response of the vessel wall to any form of insult, including physical trauma 🚚, chemical toxins ☠️ (e.g., cigarette smoke 🚬), inflammation 🔥, or hemodynamic stress 🌊.

🔧 Healing Mechanism: When the vessel wall is injured, smooth muscle cells (SMCs) or their precursors migrate from the media into the intima.

🧬 Neointima Formation: These cells proliferate and synthesize extracellular matrix (ECM), forming a "neointima" that fills the area of injury, much like a scar 🩹.

🚧 Outcome: While intended to repair the vessel, persistent or recurrent insults lead to further thickening, which results in permanent narrowing (stenosis) of the vessel lumen.

II. The Primary Responder: Endothelial Cells (ECs) 📌

The endothelium is not a passive lining but a highly dynamic, metabolically active organ system that regulates vascular tone ⚖️, coagulation 📌, and inflammation 🔥.

A. Basal vs. Activated States ⚡

🟢 Basal State: Promoted by normal blood pressure and laminar flow 🌊, maintaining a

non-adhesive, non-thrombogenic surface.

🔴 **Activated State:** Triggered by injurious stimuli (e.g., turbulent flow 🌀, cytokines 💉, lipids 🍔, or hypertension 📈), leading to the expression of pro-coagulant and pro-inflammatory factors.

B. Endothelial Dysfunction (ED) ⚠️

Prolonged exposure to high levels of stimuli leads to endothelial dysfunction, characterized by:

⬇️ Impaired vasodilation due to decreased nitric oxide (NO).

🏠 Increased permeability and leukocyte adhesion.

🩸 Hypercoagulable states that can initiate thrombosis.

III. The Executive Role: Smooth Muscle Cells (SMCs) 🏗️

SMCs are responsible for the structural integrity of the vessel and undergo significant phenotypic changes during the response to injury.

A. SMC Activation 🔥

In response to injury, SMCs are stimulated by factors such as platelet-derived growth factor (PDGF) 💉 and inflammatory mediators like interferon- γ (IFN- γ) 🔥.

🔄 **Phenotypic Switch:** Medial SMCs, which are normally "sleepy" 😴 and contractile, transform into neointimal SMCs that are non-contractile but have high proliferative and synthetic capacity 🧬.

🧱 **ECM Production:** These activated SMCs produce collagen 🧶, elastin 🧵, and proteoglycans 🧴 to rebuild the vessel wall.

ARTERIOSCLEROSIS ✨ ✨

Arteriosclerosis is a broad medical term for "hardening of the arteries". It happens when the blood vessels that carry oxygen and nutrients from your heart to the rest of your body become thick, stiff, and lose their natural elasticity. This hardening restricts blood flow to your organs and tissues.

TYPES OF ARTERIOSCLEROSIS

1. ATHEROSCLEROSIS

Atherosclerosis (from the Greek roots for "gruel" 🍲 and "hardening" 🪨) is a specific, clinically significant pattern of arteriosclerosis (the generic term for "hardening of the arteries"). It is characterized by intimal lesions called atheromas (atherosclerotic plaques) that protrude into vascular lumina and can rupture 💣, causing sudden, catastrophic occlusions 🚫🩸.

I. Conceptual Framework:

● The Response-to-Injury Hypothesis 🧠

This is the cornerstone for understanding atherosclerosis. It views the disease as a chronic inflammatory response 🔥 of the arterial wall to endothelial injury 🚚. The progression is not a passive accumulation of "sludge" 🗑️ but a highly active, dynamic interplay of modified lipoproteins 🧬, macrophages 🦠, T lymphocytes ✂️, and vascular wall cells .

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II. Pathogenesis: Hierarchical Steps 📅

🚫 Endothelial Injury & Dysfunction: Chronic insults (e.g., hyperlipidemia 🍔, hypertension 📈, toxins from smoking 🚬) lead to endothelial dysfunction, characterized by increased permeability 🏠 and leukocyte adhesion 🩸.

👉 Lipid Accumulation: Low-density lipoprotein (LDL) accumulates in the intima.

🔥 Oxidation & Inflammation: Lipids are oxidized by free radicals generated by macrophages 🦠 or endothelial cells 🚚. Oxidized LDL is cytotoxic and attracts monocytes, which migrate into the intima and differentiate into macrophages.

👉 Foam Cell Formation: Macrophages engulf oxidized LDL via scavenger receptors, becoming foam cells.

🔔 Smooth Muscle Cell (SMC) Recruitment: Factors released from platelets 🩸 and macrophages (like PDGF 🧬) trigger SMCs to migrate from the media into the intima.

🧱 Plaque Maturation: These SMCs proliferate and synthesize extracellular matrix (collagen 🧬), transforming a simple fatty streak into a mature, fibrous atheromatous plaque.

III. Morphology and Plaque Anatomy 🧫

🟡 Fatty Streaks: The earliest visible lesions; they are flat, yellow streaks composed of foamy macrophages 🦠. They are found in almost all children over 10 years old 🧒 and do not significantly disturb blood flow .

📌 Mature Plaque: Consists of two main parts:

📌 Fibrous Cap: A superficial layer of SMCs, macrophages 🦠, T cells ✂️, and dense collagen 🧬.

📌 Necrotic Core: A soft, grumous center containing cholesterol crystals 💎, lipid-laden foam cells 👉, calcium 🪨, and necrotic debris ☠️.

🪨 Calcification: Atheromas often undergo secondary calcification ("hardening") over time ⌚.

IV. Clinical Application & Consequences 🏥

The pathology manifests depending on the size of the vessel and the stability of the plaque:

🚧 Atherosclerotic Stenosis: Gradual enlargement of the plaque. Critical stenosis occurs when the vessel is roughly 70% occluded 📉, at which point blood flow cannot meet tissue demand during exertion 🏃 (e.g., stable angina ❤️ or intermittent claudication 🦵).

💧 Acute Plaque Change: A "vulnerable plaque" (thin cap 📄, large lipid core 🧈, many inflammatory cells 🔥) is prone to rupture or erosion. This exposes highly thrombogenic material ⚠️, triggering thrombosis 🩸 and sudden occlusion 🚫 (e.g., Myocardial Infarction 🍷 or Stroke 🧠).

💧 Aneurysm Formation: Plaque-induced pressure or ischemia causes atrophy of the underlying media. This weakening leads to dilation and potential rupture 💣 of the vessel wall.

V. Major Risk Factors (Multiplicative Effect) ⚠️

Risk factors often work together to accelerate the disease 🏃:

🔄 Modifiable: Hyperlipidemia 🧈 (high LDL, low HDL), Hypertension 📈 (the major cause of LVH ❤️), Smoking 🚬, Diabetes 🩺, and Inflammation 🔥 (measured by C-reactive protein 🧪).

🧬 Non-modifiable: Increasing age 🧓, male gender 🧑 (premenopausal women are relatively protected), and Family History 🧑🧑 (the most important independent risk factor).

💊 Statins: Drugs that inhibit HMG-CoA reductase ⚙️ to lower cholesterol 🧈 and help stabilize plaques by reducing inflammation 🔥.

🫀 The clinical effects of atherosclerosis are primarily driven by two pathophysiologic mechanisms: narrowing or complete obstruction of vessel lumina (which can occur gradually via stenosis or acutely via thrombosis/embolism ⚠️) and weakening of vessel walls 🧱⬇️, which leads to dilation 💧 and potential rupture 💣. While atheromatous plaques often remain clinically silent 😊 for decades ⏳, they typically manifest in middle age or later 🧓 when they reach a critical threshold of obstruction.

1. Cardiac Consequences (Ischemic Heart Disease) ❤️👑

Atherosclerosis is the underlying cause of more than 90% of cases of Ischemic Heart Disease (IHD) 🫀.

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Stable Angina: Gradual plaque enlargement leads to critical stenosis 🚧, defined as approximately 70% occlusion of the vessel lumen. At this stage, chronic hypoperfusion 🩸⬇️ is sufficient for resting needs 🛌, but exertion 🏃 triggers chest pain because demand exceeds supply ⚖️.

🍷 Acute Coronary Syndromes: Sudden disruption 💣 (rupture, erosion, or ulceration) of a

plaque triggers acute thrombosis 🩸, leading to partial or complete vascular obstruction 🚫.

🔥 Unstable Angina: Characterized by increasingly frequent pain 😞, often at rest 🛌, caused by non-occlusive or transient thrombi 🩸.

Myocardial Infarction (MI): Complete thrombotic occlusion 🚫 leads to necrosis (death 💀) of the heart muscle 🍖.

🔥📉 Chronic IHD with Heart Failure: Progressive cardiac decompensation 📉 resulting from accumulated small ischemic insults ⚡ or prior infarctions eventually leads to mechanical pump failure ❌.

⚡💀 Sudden Cardiac Death (SCD): Often the first manifestation of atherosclerosis, typically resulting from a lethal arrhythmia 🔄 (such as ventricular fibrillation ⚡) triggered by myocardial irritability from ischemia 🩸📉.

2. Cerebral Consequences (Cerebrovascular Disease) 🧠

Atherosclerosis in the carotid arteries or circle of Willis 🔄 leads to significant neurological deficits 🧠.

💥 Stroke (Cerebral Infarction): Acute occlusion 🚫 or embolization 🩸 from carotid plaques can cause permanent brain damage.

🧠 Transient Ischemic Attacks (TIAs): Brief, reversible 🔄 episodes of neurologic dysfunction ⚡ that serve as prodromes 🔥 for impending strokes.

📉 Ischemic Encephalopathy: Chronic arterial hypoperfusion 🩸📉 can lead to global cognitive decline.

3. Peripheral and Visceral Consequences 🦵👤

🦵🚧 Peripheral Arterial Disease (PAD): Primarily affects the lower extremities 🦵 (e.g., the popliteal artery).

🦵🦶 Intermittent Claudication: Predictable ischemic leg pain 😞 that occurs during exercise 🏃 and is relieved by rest 🛌.

🦵👤 Gangrene: Severe chronic narrowing or sudden occlusion 🚫 can lead to tissue death 💀 in the extremities, a complication that is 100-fold more common in diabetic patients 🩸.

🩺📉 Renovascular Hypertension: Atherosclerotic narrowing 🚧 at the ostia of the renal arteries reduces blood flow 🩸📉 to the kidneys 🧠, triggering the renin-angiotensin system ⚙️ and causing systemic high blood pressure 📈.

🩺🩸📉 Bowel Ischemia: Chronic hypoperfusion of the mesenteric arteries 👤 leads to intestinal distress ⚠️.

4. Structural and Embolic Complications

📍 Aneurysms: Plaque-induced pressure ⚡ or ischemia 🩸📉 causes atrophy of the underlying media and loss of elastic tissue 🛠️❌. This leads to structural weakening and aneurysmal dilation 📍, most commonly in the infrarenal abdominal aorta (AAA). These are prone to fatal rupture 💀💥.

Mural Thrombosis: Clots 🩸 can form over the surface of a disrupted plaque 💥 or within an aneurysm sac 📍, potentially breaking off as emboli that occlude distal vessels 🚫, such as the popliteal artery 🦵.

💎 Atheroembolism: A ruptured plaque 💥 can discharge debris 🌿, such as cholesterol

crystals 💎, into the bloodstream 🩸, which lodge in small arterial branches 🌿.

🫀 Calcification: Over time ⌚, mature atheromas undergo secondary calcification 🪨, which further "hardens" the arteries and reduces their elasticity ↺↓.

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2) MÖNCKEBERG MEDIAL CALCIFIC SCLEROSIS

🪨 Mönckeberg medial calcific sclerosis is one of the four distinct patterns of arteriosclerosis 🫀 (a general term for "hardening of the arteries"). While other forms like atherosclerosis or arteriolosclerosis can cause significant illness 😟, Mönckeberg sclerosis is generally considered a benign condition.

1. Pathological Features 🔬

📍 Location: It involves the tunica media 🧱 of small to medium-sized muscular arteries, such as the radial 🖐️ or uterine 🍑 arteries.

🪨 Calcification: The hallmark ⭐ of the disease is the presence of calcific deposits 🪨, which are typically centered on the internal elastic lamina.

🩸 Lumen Patency: Crucially, these lesions do not encroach on the vessel lumen 🚫. Because the interior of the vessel remains open 🟢 and blood flow is not obstructed 🌊, the condition does not typically cause downstream ischemia ❌.

2. Clinical Significance 🏥

📌 Incidental Finding: Because it is usually asymptomatic 😊, it is most often discovered as an incidental observation 🔍 on X-rays 📷, where the calcified arteries become visible 🪨.

📌 Benign Nature: It is a clinically insignificant process 🕒 that generally carries no serious sequelae 🚫⚠️.

📌 Demographics: It is most commonly found in individuals older than 50 years of age 🎂.

📌 Associations: While often a product of aging ⌚, it can be associated with diabetes mellitus 🍷 and chronic kidney disease 🩸⚠️.

3. Differential Diagnosis ⚖️

It is important to distinguish Mönckeberg sclerosis from other forms of arterial hardening 🫀:

Atherosclerosis: Involves the intima (not the media), creates plaques 🍌 that narrow the lumen 🚫, and is clinically significant ⚠️.

Arteriolosclerosis: Affects small arteries and arterioles 🩸, causes luminal narrowing, and is associated with hypertension 📈 and diabetes 🍷.

Fibromuscular Intimal Hyperplasia: A non-atherosclerotic process 🚫📄 driven by inflammation 🔥 or mechanical injury ⚙️ (like stenting 🩺).

3) ARTERIOLOSCLEROSIS

🩸 Arteriolosclerosis is a specific pattern of arteriosclerosis (the general "hardening of the arteries") that involves small arteries and arterioles. It is fundamentally related to hypertension 📈 and is a critical regulator of blood pressure ⚖️, as even small changes in arteriolar lumen

size 📏 have profound effects on vascular resistance .

⚡ There are two primary variants of arteriolosclerosis, distinguished by their triggers ⚠️ and morphological appearances 🔍:

1. Hyaline Arteriolosclerosis 🏠

This type is most commonly associated with benign hypertension 📈 and diabetes mellitus 🩺.

🔬 Morphology: It is characterized by homogeneous, pink ❤️, hyaline thickening of the arteriolar walls. This "glassy" ⚡ appearance results in the loss of underlying structural detail 🔍❌ and significant luminal narrowing .

⚙️ Pathogenesis: The lesions arise from the leakage 💧 of plasma components across injured endothelial cells 🚚 into the vessel walls, combined with increased extracellular matrix (ECM) 🕸 production by smooth muscle cells (SMCs) 🏗 in response to chronic hemodynamic stress.

🏠 Clinical Significance: In the kidneys 🩸, the resulting vascular compromise ⚠️ leads to nephrosclerosis (glomerular scarring).

In diabetic patients 🩺, the underlying cause is often hyperglycemia-associated endothelial dysfunction. It can also cause damage to the retina 👁.

2. Hyperplastic Arteriolosclerosis 🍅🩸

This variant is characteristic of severe or malignant hypertension 🚨📈.

🔬 Morphology: Vessels exhibit a concentric, laminated thickening 🌀 known as "onion-skinning" 🍅, which causes severe luminal narrowing or total obliteration.

⚙️ Pathogenesis: The laminations are composed of proliferating smooth muscle cells and a thickened, reduplicated basement membrane 🧱.

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🏠 Clinical Significance: In the context of malignant hypertension 🚨 (where blood pressure often exceeds 200/120 mm Hg), these changes are frequently accompanied by fibrinoid deposits 🩸 and vessel wall necrosis 💀, a condition termed necrotizing arteriolitis 🚚💥. This is particularly prominent in the kidneys 🩸 and can lead to rapid organ failure ⚡❌.

VASCULITIS

💥 Vasculitis is a general term for vessel wall inflammation 🔥🧱. This condition is a heterogeneous group of disorders ⚠️ that can lead to systemic manifestations 🌡️😓 (fever, malaise, myalgia 💪😓) and significant organ dysfunction 🩺🧠🩸 depending on the specific vascular bed affected.

I. Pathogenesis: Mechanisms of Vascular Injury

The two most common pathogenic mechanisms of vasculitis are immune-mediated 🛡️ (noninfectious) inflammation 🔥 and direct vascular invasion 🦠 by infectious pathogens. It is clinically critical ⚠️ to distinguish between these two because immunosuppressive therapy 💊 for the former could dangerously exacerbate 📈 the latter.

1. Noninfectious (Immune-Mediated) Mechanisms

Most vasculitides have an immunologic basis 🧬. Four primary mechanisms are involved:

🔧 Immune Complex Deposition:

Antibodies react with antigens 🧬 (either foreign 🌐, like drugs 💊 or microbes 🦠, or self 🧑, like in SLE) to form complexes that deposit in the vessel wall 🧱. This is seen in SLE vasculitis, Henoch-Schönlein Purpura (IgA vasculitis), and roughly 30% of Polyarteritis Nodosa cases associated with Hepatitis B.

🔪🩸 Anti-Neutrophil Cytoplasmic Antibodies (ANCA): These autoantibodies react with enzymes in neutrophil granules 🧴 (e.g., PR3 or MPO). They can directly activate neutrophils 🌟, causing them to release reactive oxygen species ⚡ and proteolytic enzymes ✂️ that injure endothelial cells (ECs). This "pauci-immune" (lacking visible antibody/complement) pattern is found in Granulomatosis with Polyangiitis (GPA) and Microscopic Polyangiitis (MPA) 🔬.

★ Anti-Endothelial Cell Antibodies:

Antibodies directed against ECs or smooth muscle cells 🏗️ can trigger inflammation 🔥, as seen in Kawasaki disease ❤️.

🧬✂️ Autoreactive T Cells (Cell-Mediated Immunity):

Some forms are driven by T-cell responses ⚡ to vessel wall antigens 🧱, often leading to granuloma formation 🟡. This is characteristic of Giant Cell Arteritis 👁️ and Takayasu Arteritis 🩸.

2. Infectious Vasculitis 🔬🔥

This occurs through direct invasion 🦠 of arteries by bacteria 🦠 or fungi 🍄 (e.g., Aspergillus or Mucor). This can result from a nearby infection 😷 (pneumonia 🫁, abscess 🩹) or hematogenous spread 🩸. It can weaken the wall 🧱⬇️, leading to mycotic aneurysms, or cause thrombosis and infarction.

TYPES OF VASCULITIS

Large-vessel vasculitis 🩸🌳 refers to a group of chronic inflammatory disorders 🔥 that predominantly affect the aorta and its major branches 🌿. The two most significant clinical entities in this category are Giant Cell (Temporal) Arteritis and Takayasu Arteritis, both of which are characterized by granulomatous inflammation 🔥.

1. Giant Cell (Temporal) Arteritis (GCA) 👁️🩸

This is the most common form of vasculitis among older adults in developed countries 🌐. It is a chronic inflammatory disorder 🔥 that typically involves the cranial arteries 🧠, specifically the temporal ⌚, vertebral, and ophthalmic 👁️ arteries, though it can also affect the aorta ❤️ (giant cell aortitis).

🧬 Pathogenesis:

It results from a T-cell-mediated immune response ✂️ to an unknown vessel wall antigen ? 🧱. Activated CD4+ T-cells ⚡ and macrophages 🦠 are central effector cells, and pro-inflammatory cytokines 🔧 like TNF play a significant role 🔥.

🔬 Morphology:

🔪 Pathologic changes are patchy/segmental along the vessel 🩸.

🔪 Characterized by granulomatous inflammation in the inner media 🧱 centered on the

internal elastic lamina 🏗️.

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📌 The infiltrate includes lymphocytes 🟡, macrophages 🦠, and multinucleate giant cells 🧟.

📌 Key histologic features include fragmentation of the internal elastic lamina ❌ and nodular intimal thickening that causes luminal narrowing 🚧.

🏥 Clinical Features:

✅ Occurs almost exclusively in patients older than 50 years 🎂.

✅ Symptoms include fever 🌡️, fatigue 😴, weight loss ⚖️⬇️, and intense facial pain or headache 🤯 along the temporal artery.

👁️ Ocular symptoms (diplopia 👁️↔️ to complete vision loss) occur in 50% of patients due to ophthalmic artery involvement; this requires urgent treatment 🚑💊 to prevent permanent blindness 🕶️.

💪 It is frequently associated with polymyalgia rheumatica 🦴😞.

2. Takayasu Arteritis ("Pulseless Disease")

Takayasu arteritis is a granulomatous vasculitis primarily involving the aorta (especially the arch 🏰) and its main branches 🌿.

🦠 Pathogenesis:

It is believed to have an autoimmune etiology. Key players include cytotoxic CD8+ 🗡️ and CD4+ T-cells ⚡, macrophages 🦠, and natural killer (NK) cells 💀.

🔬 Morphology:

🩸 The inflammation starts at the vasa vasorum and progresses to transmural fibrosis.

✳️ In the active stage, it involves intense mononuclear inflammation and giant cells 🧟 with patchy medial necrosis 💀.

✳️ Chronic cases show fibroblast proliferation and scarring 🩹, resulting in a "tree-bark" 🌳 appearance of the intimal surface.

✳️ The result is a classic triad of stenosis, occlusion, and aneurysm formation 🎈 in the great vessels 🫀.

🏥 Clinical Features:

👩 Primarily affects young women 🧑 (typically younger than 50 🎂).

💪🚫 Characterized by weakening of pulses in the upper extremities and variations in blood pressure 📈 between limbs.

😞 Manifests with ocular disturbances 👁️, neurologic deficits 🧠⚠️, and sometimes systemic hypertension 📈 if the renal arteries 🩸 are involved.

Diagnosis of Large-Vessel Vasculitis 🫀🔥

Diagnosis requires clinicopathologic correlation 🧪🩸.

🧪 Serology: Nonspecific markers ⚠️ like a markedly elevated 📈 Erythrocyte Sedimentation Rate (ESR) are common.

 Imaging: Angiography  or PET-CT  can identify stenosis , occlusion , or vessel wall inflammation .

 Tissue Biopsy: The gold standard  for diagnosis; for GCA , a biopsy must be taken from an actively involved segment of the temporal artery.

MEDIUM-VESSEL VASCULITIS

Medium-vessel vasculitis is a category of vessel wall inflammation  that predominantly targets medium-sized muscular arteries . The two primary clinical entities in this group are Polyarteritis Nodosa (PAN)  and Kawasaki Disease . Unlike large-vessel vasculitis, these disorders are generally not granulomatous, and unlike small-vessel vasculitis, they typically spare the smallest vessels like capillaries and venules.

1. Polyarteritis Nodosa (PAN)

PAN is a systemic necrotizing vasculitis  of small- or medium-sized muscular arteries . It is a life-threatening condition if left untreated.

 Pathogenesis

 Hepatitis B Association

Approximately 30% of cases are associated with chronic Hepatitis B infection. In these patients, immune complexes composed of Hepatitis B surface antigen (HBsAg) and antibodies deposit in the vessel walls, triggering inflammation .

 ANCA Status

PAN is generally not associated with ANCAs (Anti-Neutrophil Cytoplasmic Antibodies), which helps distinguish it from MPA.

 Morphology

 Segmental Transmural Necrosis

The inflammation is sharply segmental and affects the full thickness of the arterial wall (transmural).

 Fibrinoid Necrosis

The acute phase is characterized by a mixed inflammatory infiltrate (neutrophils and mononuclear cells) and fibrinoid necrosis.

 Temporal Heterogeneity 

A hallmark of PAN is that lesions at different stages—acute inflammation, healing, and fibrous scarring—can coexist within the same or different vessels.

 Aneurysm Formation

The inflammatory process weakens the arterial wall, leading to small microaneurysms and potential rupture.

 Organ Involvement

Typically affects:

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 Kidneys

 Heart

 Liver

 Gastrointestinal tract

 Pulmonary Sparing

A key diagnostic feature is that PAN spares the pulmonary circulation.

 Clinical Features

Commonly seen in young to middle-aged adults.

Symptoms depend on the affected vascular bed and include:

Abdominal pain

Melena

Renal failure

Muscle weakness

 Diagnosis

 Serology

HBV antigen testing

ANCA usually negative

 Imaging

Angiography is used to identify:

Microaneurysms 

Focal stenoses

Particularly in:

Mesenteric vessels

Renal vessels

 Tissue Biopsy

The gold standard for PAN, ideally taken from an actively involved site such as the kidney or skin to observe transmural inflammation.

2. Kawasaki Disease

Also known as mucocutaneous lymph node syndrome, Kawasaki disease is an acute, febrile, self-limited illness of infancy and childhood.

 Epidemiology

Approximately 80% of patients are younger than 4 years of age.

 Pathogenesis

It is believed to result from a delayed-type hypersensitivity response to an unknown infectious (likely viral) or environmental trigger in genetically susceptible individuals.

This leads to the production of auto-antibodies to endothelial cells and smooth muscle cells, which precipitate the vasculitis 🔥.

Clinical Features

Characterized by:

-  Conjunctival erythema
-  Oral erythema (blistering)
-  Edema and erythema of the hands
-  Edema and erythema of the feet
-  Desquamative rash
-  Cervical lymph node enlargement
-  Cardiovascular Significance
-  Coronary Artery Involvement

The primary clinical concern is coronary artery involvement, which occurs in roughly 20% of untreated cases.

It can lead to:

Coronary artery ectasia

Large aneurysms (7–8 mm)

Rupture

Thrombosis

Myocardial infarction

Sudden death

 Treatment

Management with:

Intravenous immunoglobulin (IVIG)

Aspirin

Markedly reduces the incidence of symptomatic coronary artery disease.

SMALL-VESSEL VASCULITIS

Small-vessel vasculitis is a category of vessel wall inflammation 🔥 that predominantly affects the smallest segments of the vascular bed, including capillaries, venules, and arterioles. These disorders are broadly divided into two groups based on their underlying immunologic mechanism:

I. ANCA-Associated (Pauci-Immune)

These forms are characterized by a "pauci-immune" pattern, meaning they typically show little to no demonstrable antibody or complement deposition in vascular lesions. They are strongly associated with Anti-Neutrophil Cytoplasmic Antibodies (ANCAs).

1. Microscopic Polyangiitis (MPA)

Pathogenesis

Primarily associated with MPO-ANCA (p-ANCA). It involves a necrotizing vasculitis without granulomatous inflammation.

★ Key Features

Necrotizing glomerulonephritis (90% of patients)

Pulmonary capillaritis

🟪 Leukocytoclastic Vasculitis

When it involves the skin, it often manifests as palpable purpura where infiltrating neutrophils undergo fragmentation (leukocytoclasia).

2. 🦠 Granulomatosis with Polyangiitis (GPA, formerly Wegener Granulomatosis)

🧬 Pathogenesis

Highly associated with PR3-ANCA (c-ANCA). It is likely initiated as a cell-mediated hypersensitivity response to inhaled antigens.

▲ Classic Triad

🔴 Necrotizing granulomas of the upper/lower respiratory tract

🔴 Necrotizing or granulomatous vasculitis of small- to medium-sized vessels

🔴 Focal necrotizing, often crescentic, glomerulonephritis

🏠 Clinical Presentation

Patients often have:

✳️ Chronic sinusitis

✳️ Nasopharyngeal ulcers

✳️ Bilateral lung nodules that may cavitate

3. 🦠 Eosinophilic Granulomatosis with Polyangiitis (EGPA, formerly Churg-Strauss)

🧬 Pathogenesis

Associated with MPO-ANCA in a minority of cases.

★ Key Features

This is a small-vessel necrotizing vasculitis classically seen in individuals with:

✅ Asthma

✅ Allergic rhinitis

✅ Peripheral eosinophilia

👉 It is characterized by granulomas and striking infiltration of vessels by eosinophils.

II. 🦠 Immune Complex–Mediated Small-Vessel Vasculitis

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These disorders result from the deposition of antigen-antibody complexes into the vessel wall, which then trigger an inflammatory response.

1. 🟪 IgA Vasculitis (Henoch-Schönlein Purpura)

Most common vasculitis in children

Often follows an upper respiratory infection

Features

🟡 Deposition of IgA-rich immune complexes in:

★ Skin

★ Gastrointestinal tract

★ Kidneys

Leading to:

🔴 Purpuric skin lesions

● Abdominal pain

● Hematuria

2. ❄️ Cryoglobulinemic Vasculitis

Associated with cryoglobulins (proteins that precipitate in the cold), often in the context of chronic Hepatitis C infection.

3. 🦋 Lupus Vasculitis

A small-vessel response occurring in patients with SLE due to deposition of nuclear antigen-antibody complexes.

4. 🫁 Goodpasture Disease

Specifically mediated by antibodies directed against basement membrane collagen in:

❄️ Pulmonary alveoli

❄️ Renal glomeruli

Leading to:

🫁 Hemoptysis

🩸 Rapidly progressive glomerulonephritis

⚙️ PATHOGENESIS HIGHLIGHTS

🔪 ANCA Activation

ANCAs can directly activate neutrophils, stimulating the release of:

Reactive oxygen species

Proteolytic enzymes

Resulting in endothelial injury.

💊 Drug Hypersensitivity

Certain drugs (e.g., penicillin) can act as haptens, triggering immune complex formation and a subsequent small-vessel vasculitis that often resolves upon removal of the drug.

🔍 DIAGNOSIS OF SMALL-VESSEL VASCULITIS

Diagnosis requires clinicopathologic correlation using:

📋 Clinical Assessment

Evaluation of:

Lung involvement

Kidney involvement

Skin involvement

🧪 Serology

ANCA

Cryoglobulins

HBV/HCV

ANA

Anti-dsDNA

Anti-GBM antibodies

 Tissue Biopsy (Gold Standard)

Must be obtained from an actively involved site such as:

 Skin

 Kidney

 Lung

To identify characteristic necrotizing or granulomatous patterns.

 ANEURYSM

 An aneurysm is defined as a permanent, abnormal dilation of a blood vessel or the heart that involves the entire wall thickness and typically exceeds the normal diameter by at least 50%.

★ These dilations occur when the structural integrity of the arterial media is compromised by alterations in smooth muscle cells or the extracellular matrix.

 I. Structural Classification

Aneurysms are categorized based on whether the vessel wall remains intact:

 True Aneurysm

Involves an attenuated but intact arterial wall or thinned ventricular wall of the heart.

All three layers (intima, media, and adventitia) are present.

Common examples include:

 Atherosclerotic aneurysms

 Congenital aneurysms

 Syphilitic aneurysms

 Ventricular aneurysms following a transmural myocardial infarction

False Aneurysm (Pseudoaneurysm)

Results from a breach in the vascular wall that leads to an extravascular hematoma that freely communicates with the intravascular space.

It is often described as a "pulsating hematoma".

Examples include:

Ventricular rupture contained by pericardial adhesions

A leak at the junction of a vascular graft and a natural artery

II. Classification by Shape

Aneurysms are also described by their macroscopic appearance:

Saccular Aneurysms

These are discrete, spherical outpouchings involving only a portion of the vessel wall.

They range from 5 to 20 cm in diameter and often contain a mural thrombus.

Fusiform Aneurysms

These consist of circumferential, symmetrical dilations of a long vascular segment, up to 20 cm in diameter.

  They frequently involve:

 The aortic arch

 Abdominal aorta

 Iliac arteries

III. Clinical Types and Specific Causes

Aneurysm formation results from various factors that weaken the vessel wall:

Abdominal Aortic Aneurysm (AAA)

The most common type, typically occurring between the renal arteries and the aortic bifurcation.

Atherosclerosis is the dominant factor, occurring more frequently in:

 Men

 Smokers

 Individuals over age 50

 Atherosclerotic plaques impede nutrient diffusion to the media, causing ischemic atrophy and the release of proteolytic enzymes that degrade the extracellular matrix.

 Thoracic Aortic Aneurysm

Most commonly associated with:

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Hypertension

Bicuspid aortic valves

Marfan syndrome

 Hypertension causes narrowing of the vasa vasorum, leading to outer media ischemia and cystic medial degeneration.

 Berry Aneurysms

Small, thin-walled outpouchings in cerebral vessels, classically at branch points in the circle of Willis.

They occur where the media is congenitally attenuated and are associated with autosomal-dominant polycystic kidney disease (ADPKD).

 Mycotic Aneurysms

Result from the weakening of a vessel wall due to infection, such as the embolization of a septic thrombus in infective endocarditis.

 Syphilitic Aneurysm

A rare complication of tertiary syphilis involving the vasa vasorum of the ascending thoracic aorta.

This causes obliterative endarteritis and subsequent ischemic injury to the media.

 Ventricular Aneurysm

A late complication of ischemic heart disease where a thinned, scarred area of the heart wall bulges outward following a large transmural myocardial infarction.

 Vasculitis-Associated

Inflammatory disorders like Takayasu arteritis, Polyarteritis Nodosa (PAN), and Kawasaki

disease can damage the vessel wall sufficiently to cause aneurysm formation.

⚙️ IV. Pathogenesis and Mechanisms

The fundamental basis for aneurysms is the weakening of the arterial media through several pathways:

🧬 Inadequate Connective Tissue Synthesis

Inherited mutations, such as those affecting TGF- β signaling or fibrillin synthesis (Marfan syndrome), lead to defective elastin and collagen production.

✂️ Excessive Connective Tissue Degradation

Proteolytic enzymes (matrix metalloproteinases) released from inflammatory cells in atherosclerotic plaques degrade the arterial wall's structural proteins.

🚫 Loss of Smooth Muscle Cells (SMCs)

Ischemia resulting from intimal thickening or hypertensive narrowing of the vasa vasorum leads to SMC loss and a change in synthetic phenotype.

🖋️ Cystic Medial Degeneration

A nonspecific histologic finding characterized by:

SMC dropout

Elastic tissue fragmentation

Accumulation of proteoglycan-rich ECM

⚠️ V. Major Clinical Complications

💥 Rupture

Often leading to catastrophic and fatal hemorrhage, with the risk increasing as the aneurysm grows (e.g., AAAs >5 cm).

🩸 Thrombosis and Embolization

Stasis of blood within the aneurysm promotes the formation of mural thrombi, which can break off and occlude distal vessels, such as the popliteal artery.

🚧 Obstruction and Impingement

Expanding aneurysms can obstruct branching vessels (causing ischemia to kidneys or legs) or compress adjacent structures like the ureters or vertebrae.

AORTIC DISSECTION

An aortic dissection is a catastrophic clinical event that occurs when pressurized blood gains entry to the arterial wall through a surface defect (typically an intimal tear) and separates the laminar planes of the media to form a blood-filled channel.

This condition is distinct from an aneurysm, although the two are often related; a dissection does not necessarily require preexisting aortic dilation.

I. Major Clinical Causes and Risk Factors

The fundamental cause of dissection is structural weakness of the vessel wall.

Hypertension

This is the most significant risk factor, present in more than 90% of cases, typically occurring in men aged 40 to 60.

Chronic hypertension causes narrowing of the vasa vasorum, which leads to medial smooth muscle cell loss and degenerative changes in the extracellular matrix.

Inherited Connective Tissue Disorders

Younger patients are often affected due to heritable conditions that result in abnormal aortic extracellular matrix, such as Marfan syndrome (defective fibrillin) and Ehlers-Danlos syndrome type IV (defective type III collagen).

Congenital Abnormalities

Specifically, a bicuspid aortic valve is a recognized risk factor.

Iatrogenic Factors

Dissections can be accidentally triggered during medical procedures, such as arterial cannulation for diagnostic catheterization or cardiopulmonary bypass.

Other Triggers

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Dissections may be associated with pregnancy-induced vascular remodeling (typically in the third trimester) or abrupt, transient spikes in blood pressure from cocaine abuse.

II. Classification and Types

Dissections are classified based on the anatomical portion of the aorta involved, which dictates the severity and management.

Type A (Proximal)

These involve the ascending aorta, regardless of whether the descending aorta is also affected.

These are the most dangerous forms with the highest associated mortality.

They often require rapid diagnosis, intensive anti-hypertensive therapy, and surgical plication.

● Type B (Distal)

These usually begin beyond the takeoff of the great vessels (distal to the subclavian artery).

Many Type B dissections can be managed conservatively with anti-hypertensive medication.

III. Clinical Manifestations and Morphology

Classic Symptom

Sudden onset of excruciating "tearing" or stabbing pain that typically begins in the anterior chest, radiates to the back between the scapulae, and moves downward as the dissection progresses.

The Intimal Tear

In most cases, the point of origin is an intimal tear in the ascending aorta within 10 cm of the aortic valve.

Path of Dissection

The blood-filled channel typically lies between the middle and outer thirds of the media.

Propagation of the dissection is actually inhibited by substantial atherosclerosis or medial scarring.

"Double-Barreled Aorta"

In some instances, the dissecting hematoma reenters the aortic lumen through a second distal intimal tear, creating a second vascular channel within the media.

IV. Major Complications

Dissections are life-threatening due to potential rupture or vessel obstruction.

Rupture

The most common cause of death is external rupture into the pericardial sac (causing fatal cardiac tamponade), pleural cavity, or peritoneal cavity.

Cardiac Involvement

Retrograde dissection can disrupt the aortic valve apparatus (causing aortic insufficiency) or

compress the coronary arteries (causing a myocardial infarction).

Vascular Obstruction

The dissection can extend into branch vessels (renal, mesenteric, iliac, or neck arteries), leading to critical organ ischemia or transverse myelitis from spinal artery compression.

BENIGN VASCULAR TUMORS AND TUMOR-LIKE CONDITIONS

Benign vascular tumors and tumor-like conditions are common lesions composed of well-organized vascular channels lined by normal-appearing endothelial cells (ECs).

I. Hemangiomas

Hemangiomas are very common, often congenital, blood-filled lesions that constitute 7% of all benign tumors of infancy and childhood.

Infantile (Strawberry) Hemangioma

This is the most common tumor of infancy.

It presents weeks after birth as a rapidly growing red lesion, undergoes a proliferative phase in the first year, and typically regresses completely by age 7.

A key diagnostic feature is that they are GLUT1 positive on immunohistochemistry (IHC).

Capillary Hemangioma

These are the most frequent type, occurring in the skin, subcutaneous tissues, mucous membranes, and internal organs such as the liver, spleen, and kidneys.

Cavernous Hemangioma

Composed of large, dilated, blood-filled vascular channels, these lesions are more infiltrative than capillary types, involve deeper structures, and do not spontaneously regress.

They can be a component of von Hippel-Lindau disease, occurring in the cerebellum, brain stem, or retina.

Pyogenic Granuloma (Lobular Capillary Hemangioma)

A reactive, rapidly growing, red pedunculated lesion that frequently ulcerates and bleeds.

A specific variant, the "pregnancy tumor," occurs in the gingiva of pregnant women and may regress after delivery.

Congenital Hemangioma

Unlike infantile types, these are present fully grown at birth and are GLUT1 negative.

They are classified as rapidly involuting (RICH) or non-involuting (NICH).

II. Lymphangiomas

These are the benign lymphatic counterparts of hemangiomas.

Simple (Capillary) Lymphangioma

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Small (1-2 cm) elevated or pedunculated lesions found predominantly in the head, neck, and axilla.

Cavernous Lymphangioma (Cystic Hygroma)

Large, unencapsulated masses (up to 15 cm) typically found in the neck or axilla of children.

These are frequently associated with Turner syndrome.

Diagnosis

Histologically, they contain proteinaceous lymph fluid rather than blood.

They are positive for lymphatic IHC markers D2-40 (podoplanin) and PROX1.

III. Glomus Tumor (Glomangioma)

Origin

These benign tumors arise from specialized smooth muscle cells (SMCs) of glomus bodies, which are arteriovenous structures involved in thermoregulation.

Clinical Features

They are exquisitely painful and most commonly found in the distal digits, particularly under the fingernails.

IV. Vascular Ectasias (Non-Neoplastic)

These are permanent dilations of preexisting small vessels and are not true neoplasms.

Nevus Flammeus

A flat "birthmark"; a special form known as the "port wine stain" grows during childhood and is associated with Sturge-Weber syndrome (mental retardation, seizures, and hemiplegia).

Spider Telangiectasia

These manifest as radial, pulsatile arrays of dilated arterioles that blanch with pressure.

They are most frequently associated with hyperestrogenic states such as pregnancy or liver

cirrhosis.

Hereditary Hemorrhagic Telangiectasia (Osler-Weber-Rendu Disease)

An autosomal dominant disorder caused by mutations in TGF- β signaling pathway genes.

Lesions can rupture, causing serious nosebleeds (epistaxis) or gastrointestinal bleeding.

MALIGNANT VASCULAR TUMORS

Malignant vascular tumors are a rare but highly aggressive group of neoplasms originating from the endothelium or supporting cells of blood and lymphatic vessels.

Their clinical behavior ranges from locally destructive (intermediate grade) to rapidly fatal with early metastasis (high grade).

I. Angiosarcoma (Malignant)

Angiosarcomas are highly malignant endothelial neoplasms that can range from well-differentiated tumors to wildly anaplastic lesions.

Clinical Settings and Causes

Skin and Soft Tissue (Elderly)

Most commonly found on the head and scalp of older adults.

Hepatic (Liver) Angiosarcoma

Specifically associated with chemical carcinogens such as vinyl chloride (used in the plastics industry), arsenic (pesticides), and Thorotrast (a radioactive contrast agent).

Stewart-Treves Syndrome

Occurs in the setting of chronic lymphedema, classically in the arm of a woman several years after a radical mastectomy for breast cancer.

Iatrogenic

Induced by prior radiation therapy or long-term indwelling foreign bodies like catheters.

Cardiac

The most common primary malignant tumor of the heart, often involving the right ventricle.

Clinical Features and Morphology

Gross

Presents as a poorly circumscribed, hemorrhagic, fleshy red-tan to gray-white mass.

Microscopic

Exhibits extreme variability.

Well-differentiated forms show irregular, anastomosing vascular channels dissecting through tissue, while poorly differentiated forms show solid sheets of pleomorphic spindle or epithelioid cells with a high mitotic rate and necrosis.

Diagnosis (IHC)

Endothelial origin is confirmed by markers such as CD31 (most specific), CD34, and von Willebrand factor.

Prognosis

Generally poor, with a 5-year survival rate of approximately 30% due to high local recurrence and early hematogenous metastasis to the lungs, liver, and bone.

II. Kaposi Sarcoma (Intermediate to Malignant)

Kaposi Sarcoma (KS) is a vascular neoplasm caused by Human Herpesvirus 8 (HHV-8), also known as Kaposi sarcoma herpesvirus (KSHV).

Clinical Types

Classic KS

Seen in older men of Mediterranean or Eastern European descent; usually remains localized to the skin of the legs.

Endemic African KS

Found in younger HIV-negative individuals in Central Africa; often very aggressive with lymph node and visceral involvement.

Transplantation-Associated KS

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Occurs in transplant recipients due to T-cell immunosuppression; may regress if immunosuppression is reduced.

AIDS-Associated (Epidemic) KS

The most common HIV-related malignancy; highly aggressive and often disseminates to the viscera and lymph nodes early.

Specific Clinical Features (Stages)

Patch Stage

Pink-to-purple macules, often on the lower extremities, showing irregular vascular channels on microscopy.

Plaque Stage

Violaceous, raised lesions with jagged dermal channels surrounded by plump spindle cells and extravasated red blood cells.

Nodular Stage

Discrete, cellular nodules of spindle cells expressing both endothelial and smooth muscle markers, featuring characteristic slit-like vascular spaces and hyaline globules.

Diagnosis (IHC)

Positive for LANA-1 (HHV-8) in the nuclei of spindle cells.

III. Epithelioid Hemangioendothelioma (Intermediate)

This is a vascular tumor of adults with behavior intermediate between benign hemangiomas and malignant angiosarcomas.

Clinical Features

Arises in association with medium- to large-sized veins in soft tissue, bone, liver, and lung.

While excision is often curative, 20% to 30% eventually metastasize.

Pathology

Characterized by cords or nests of epithelioid endothelial cells in a myxohyaline stroma.

Contains pathognomonic "blister cells" (intracytoplasmic vacuoles that may contain red blood cells).

Genetic Hallmark

A WWTR1::CAMTA1 gene fusion is present in 90% of cases.

VENOUS PATHOLOGY

I. Structural and Functional Context

Compared to arteries, veins are distensible, thin-walled vessels with high capacitance, normally containing approximately two-thirds of the total blood volume.

Adaptations

Veins have larger diameters and thinner walls with less distinct layers to accommodate low pressures.

Valves

In the lower extremities, backflow is prevented by venous valves.

Vulnerability

Because of their structure, veins are more prone to dilation, external compression, and penetration by tumors or inflammatory processes.

II. Varicose Veins

Varicose veins are abnormally dilated, tortuous veins resulting from chronically increased intraluminal pressure and weakened vessel wall support.

Superficial Leg Veins

Most common in the saphenous system.

Risk factors include obesity, pregnancy (compression of the IVC), and a familial tendency.

Clinical Features

Leads to venous valve incompetence, stasis, congestion, edema, and pain.

Complications

Can cause stasis dermatitis and chronic skin ulcerations.

Notably, embolism from these superficial veins is very rare compared to deep vein thrombosis.

Other Sites

Esophageal Varices

Result from portal hypertension (often due to liver cirrhosis); these are prone to catastrophic, fatal rupture.

Hemorrhoids

Varicose dilations of the anorectal venous plexus due to prolonged pelvic congestion (pregnancy, straining); they cause bleeding and painful ulceration.

Caput Medusae

Dilation of periumbilical veins on the abdominal wall due to portal hypertension.

III. Thrombophlebitis and Phlebothrombosis

These terms describe venous thrombosis accompanied by inflammation; more than 90% of cases occur in the deep leg veins.

Deep Vein Thrombosis (DVT)

Major Risk Factor

Prolonged vascular stasis due to immobilization (bed rest or long-distance travel).

Other Risks

Postoperative state, congestive heart failure, pregnancy, oral contraceptives, and malignancy.

Trousseau Syndrome

Also known as "migratory thrombophlebitis," where procoagulant factors from a malignancy cause evanescent venous thrombi at different sites.

The Main Danger

DVT is the primary source of pulmonary thromboembolism, a serious and potentially fatal complication.

Clinical Signs

May include edema, cyanosis, heat, and pain (e.g., the Homan sign), but many cases are asymptomatic, especially in bedridden patients.

IV. Vena Cava Syndromes

Superior Vena Cava Syndrome

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Usually caused by neoplasms (e.g., bronchogenic carcinoma or lymphoma) that compress or invade the SVC.

It results in marked dilation of head, neck, and arm veins, often with cyanosis and respiratory distress.

Inferior Vena Cava Syndrome

Caused by compression from neoplasms or a propagating thrombus.

Certain tumors (e.g., renal cell carcinoma) have a tendency to grow within the vein as a continuous column.

It leads to marked lower-extremity edema and abdominal collateral vein distention.

V. Systemic Manifestations of Venous Congestion

Chronic passive congestion, often resulting from right-sided heart failure, has profound effects on visceral organs:

Nutmeg Liver

Intense centrilobular congestion where congested red areas alternate with paler parenchyma.

Congestive Splenomegaly

Elevated pressure in the portal vein leads to a tense, enlarged spleen.

Ascites

Accumulation of transudate in the peritoneal cavity due to systemic venous engorgement.

VI. Primary and Reactive Pathologies

Raynaud Phenomenon

Exaggerated vasoconstriction of digital arteries and arterioles in response to cold or emotion, causing a "white-blue-red" color change sequence.

Arteriovenous (AV) Fistulas

Abnormal direct connections between arteries and veins that bypass capillaries; large ones can lead to high-output cardiac failure.

Masson Tumor (Papillary Endothelial Hyperplasia)

A reactive, organizing thrombus (often in a vein) that microscopically mimics a vascular tumor like angiosarcoma.

LYMPHATIC PATHOLOGY

Primary disorders of the lymphatics are uncommon, they are frequently involved secondarily by infectious, inflammatory, or malignant processes.

I. Acute Inflammation: Lymphangitis

Lymphangitis refers to the acute inflammation of lymphatic channels, typically triggered by the drainage of bacteria from a site of distal infection.

Clinical Causes

The most common cause is Group A β -hemolytic streptococci (specifically *Streptococcus pyogenes*).

Features

Red Streaks

Clinically manifests as painful, erythematous subcutaneous streaks extending from the initial infection site toward regional lymph nodes.

Lymphadenitis

The draining lymph nodes often become enlarged and tender (acute lymphadenitis).

Risk

If the immune system fails to contain the bacteria within the lymph nodes, they can enter the venous circulation, potentially leading to bacteremia or sepsis.

II. Impaired Drainage: Lymphedema

Lymphedema is the accumulation of protein-rich interstitial fluid resulting from impaired lymphatic drainage.

This is categorized as either primary or secondary.

1. Primary (Congenital) Lymphedema

This is rare and results from developmental abnormalities such as lymphatic agenesis or hypoplasia.

An example is Milroy disease (hereditary familial congenital lymphedema).

2. Secondary (Acquired/Obstructive) Lymphedema

This is much more common and stems from the obstruction of previously normal lymphatic channels.

Clinical Causes

Malignancy

Tumors compressing or invading lymphatic channels or regional nodes.

Surgery

Surgical removal of lymph nodes (e.g., axillary node dissection during a radical mastectomy for breast cancer).

Filariasis

Chronic parasitic infection by *Wuchereria bancrofti*, which is a major global cause.

Inflammation/Radiation

Scarring from radiation therapy or post-inflammatory thrombosis.

Features and Consequences

Hydrostatic Pressure

Obstruction increases pressure in lymphatics distal to the site, forcing fluid into the tissues.

Tissue Changes

Chronic stagnant lymph leads to impaired immunity and dense dermal fibrosis.

The skin may develop a thickened, "orange peel" appearance known as peau d'orange.

Elephantiasis

An extreme form of chronic lymphedema characterized by massive limb enlargement and extensive fibrosis.

Chylous Effusions

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Rupture of obstructed, dilated lymphatics can leak milky lymph into body cavities, causing chylous ascites (abdomen), chylothorax (pleura), or chylopericardium.

III. Lymphatic Neoplasms and Malformations

1. Benign: Lymphangioma

These are benign congenital malformations of lymphatic channels, distinguished from hemangiomas by the absence of red blood cells within the channels.

Simple (Capillary) Lymphangioma

Small (1-2 cm) elevated or pedunculated lesions, most common in the head and neck.

Cavernous Lymphangioma (Cystic Hygroma)

Large (up to 15 cm), unencapsulated, disfiguring masses typically found in the neck or axilla of children.

These are frequently associated with Turner syndrome.

2. Malignant: Lymphangiosarcoma

This is a rare and highly aggressive endothelial malignancy.

Clinical Setting

It most often arises in the setting of long-standing chronic lymphedema.

Stewart-Treves Syndrome

A specific clinical scenario where lymphangiosarcoma develops in a lymphedematous arm years after a radical mastectomy.

IV. Summary of Diagnostic Markers

To distinguish lymphatic tumors from other vascular neoplasms, specific immunohistochemical (IHC) markers are used:

D2-40 (podoplanin)

A highly specific marker for lymphatic endothelium.

PROX1

Another key marker for identifying lymphatic origin.

Morphology

Histologically, lymphatic channels are thin-walled and contain clear, proteinaceous lymph fluid rather than blood.

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HEART PATHOLOGY

HEART FAILURE (HF) / CONGESTIVE HEART FAILURE (CHF)

Heart failure (HF), also known as congestive heart failure (CHF), is a clinical syndrome that occurs when the heart is unable to pump blood at a rate sufficient to meet the metabolic requirements of tissues, or can only do so at abnormally high filling pressures.

Key Definition:

The heart cannot effectively pump blood to meet the body's needs, resulting in inadequate tissue perfusion and/or congestion.

Important Note:

HF is the common end point of many cardiac diseases and is a progressive condition associated with a poor prognosis.

I. Etiology and Common Causes

Heart failure usually develops gradually due to chronic cardiac stress or progressive loss of

functioning myocardium.

- ♦ Primary Causes

-  Coronary Artery Disease (CAD)

-  Hypertension (HTN)

-  Diabetes Mellitus (DM)

These are the most common underlying causes of heart failure.

- ♦ Left-Sided Heart Failure Causes

-  Ischemic Heart Disease (IHD)

-  Systemic Hypertension

-  Mitral Valve Disease

-  Aortic Valve Disease

-  Primary Myocardial Diseases (e.g., amyloidosis)

- ♦ Right-Sided Heart Failure Causes

-  Most commonly caused by left-sided heart failure

-  Chronic lung diseases (Cor Pulmonale)

-  Pulmonary Thromboembolism

-  Tricuspid Valve Disease

-  Pulmonic Valve Disease

- ♦ High-Output Heart Failure

Occurs when cardiac output is insufficient despite increased pumping activity.

Examples:

-  Hyperthyroidism (increased tissue demand)

-  Severe Anemia (decreased oxygen-carrying capacity)

- ♦ Exacerbating Factors

-  Medication noncompliance

-  Excess salt intake

 Excess water intake

 Pulmonary infections

 Anemia

 Negative inotropic drugs (e.g., some calcium channel blockers and β -blockers)

II. Fundamental Mechanisms of Dysfunction

Heart failure develops through one or more major mechanisms.

♦ Systolic Dysfunction (Pump Failure)

 Reduced myocardial contractility

 Decreased ejection of blood

Common causes:

 Ischemic Heart Disease

 Hypertension

 Valvular Heart Disease

♦ Diastolic Dysfunction (Filling Failure)

 Stiff and noncompliant ventricle

 Reduced ventricular filling

Causes:

 Massive Left Ventricular Hypertrophy

 Myocardial Fibrosis

 Amyloid Deposition

 Constrictive Pericarditis

♦ Forward Failure

 Inadequate cardiac output

 Reduced tissue perfusion

 Fatigue

 Cold extremities

 Dizziness

- ◆ Backward Failure

 Blood accumulates behind the failing ventricle

 Venous congestion develops

 Pulmonary edema (left HF)

 Peripheral edema (right HF)

III. Mechanisms of Adaptation (Compensatory Responses)

When cardiac output falls, the body activates compensatory mechanisms to maintain circulation.

① Frank-Starling Mechanism

 End-diastolic filling volume

 Ventricular dilation

 Myocardial fiber stretching

 Stronger contraction

 Increased cardiac output

 Disadvantage:

 Increased wall tension

 Increased oxygen demand

 Eventually worsens heart failure

② Neurohumoral Activation

- ◆ Sympathetic Nervous System (SNS)

 Releases Norepinephrine

Effects:

 Increased heart rate (HR)

 Increased contractility

 Increased peripheral vascular resistance

- ◆ Renin-Angiotensin-Aldosterone System (RAAS)

🩸 Reduced renal perfusion stimulates renin release

Sequence:

Renin ➡ Angiotensin I ➡ Angiotensin II ➡ Aldosterone

Effects:

🩸 Vasoconstriction

🧂 Sodium retention

💧 Water retention

⬆ Increased preload

⬆ Increased blood volume

◆ Atrial Natriuretic Peptide (ANP)

❤ Released from stretched atria

Functions:

🚽 Promotes diuresis

🧂 Increases sodium excretion

🩸 Causes vasodilation

⚖ Counteracts RAAS activation

③ 💪 Myocardial Hypertrophy

Cardiac muscle cells enlarge to adapt to increased workload.

◆ Pressure Overload

Examples:

⬆ Hypertension

🧱 Aortic Stenosis

Mechanism:

🧱 Sarcomeres added in parallel

Result:

🔴 Concentric Hypertrophy

 Increased wall thickness

 No major chamber enlargement

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♦ Volume Overload

Examples:

 Mitral Regurgitation

 Aortic Regurgitation

Mechanism:

 Sarcomeres added in series

Result:

 Ventricular Dilation

 Enlarged chamber size

 Progression to Decompensated Heart Failure

Initially, compensatory mechanisms help maintain cardiac output (Compensated HF). 😊

However, prolonged activation becomes harmful:

 Progressive myocardial damage

 Ventricular remodeling

 Fluid overload

 Pulmonary congestion

 Peripheral edema

 Further decline in cardiac function

➡ This stage is called Decompensated Heart Failure, where compensatory mechanisms can no longer maintain adequate circulation.

 Quick Summary

 HF = Inability of the heart to pump sufficient blood.

 Major causes = CAD, HTN, DM, valvular disease.

💔 Systolic HF = Impaired contraction.

🧱 Diastolic HF = Impaired filling.

➡ Forward failure = Poor tissue perfusion.

⬅ Backward failure = Venous congestion.

★ Compensatory mechanisms:

Frank-Starling mechanism

SNS activation

RAAS activation

ANP release

Myocardial hypertrophy

⚠ Long-term compensation eventually causes worsening heart failure and decompensation.